Deep Learning-Based Classification of Drinking Water Waste Using Satellite and Visual Data for Sustainable Environmental Design

Omar Khaled Masoud, Marco Makram Barty, Shaker El-Sappagh

Computer Science and Engineering, Galala University, Suez, Egypt

Abstract. You Urbanization and water pollution problems are on the rise, necessitating new solutions for environmental sustainability in waste management. In this study, a deep learning model of drinking water waste classification using satellite and visual data is proposed to support environmental sustainability. Waste types such as organic wastes, excess treatment chemicals, and microplastics originating from domestic, commercial, and institutional sources are categorized by an optimized convolutional neural network model using the Bayesian hyperparameter tuning technique. To enhance accuracy, multi-source images and preprocessing methods such as augmentation and normalization are employed by the model, Preliminary results show improved performance over existing models, with Bayesian optimization being used for improved efficiency. The solution based on AI presents a replicable model in waste management, pollution, and green urban planning, leading the way towards smart environmental management.
Keywords—Bayesian optimization, deep learning, drinking water waste classification, satellite imagery, sustainable environmental design.

1 Introduction

Effective water drinking waste control has become a major business and environmental issue in the era of rapid urbanization and industrialization. Water treatment plants, municipalities, and environmental agencies have been increasingly burdened with monitoring and classifying various kinds of waste that contaminate drinking water resources [1, 2]. Traditional methods for such classification, such as manual sampling and chemical testing, not only take a lot of time and resources but also fail to scale with the ever-growing demand for environmental monitoring in real-time. Recent advancements in artificial intelligence (AI) in the form of deep learning and computer vision have shown revolutionary potential to facilitate the automation of complex environmental monitoring processes [2, 3]. Deep Convolutional Neural Networks (CNNs) have been extremely effective for image-based classification problems in areas [4]. Deep learning algorithms for the scenario of drinking water waste can process and analyze satellite- and ground-level high-resolution images to recognize and classify waste such as organic residues, microplastics, and leftover treatment chemicals [5-9]. Such capabilities make it possible to significantly improve classification accuracy, processing time, and operating scalability.

Transfer learning has also enhanced the transferability of deep learning models to environmental monitoring [15]. By using pretrained models such as DenseNet121, InceptionV3, and VGG19, it is possible for researchers to fine-tune models previously trained on large-scale datasets (e.g., ImageNet) to domain-specific issues with limited labeled data. But the majority of existing implementations are susceptible to performance deterioration in real-world field deployments due to unsuitable model optimization, environmental variations, and ungeneralizability to the unique visual character of drinking water waste. Moreover, heuristic or manual model hyperparameter adjustment is likely to yield unfriendly and unstable performance, preventing the deployment of reliable AI-based systems in real working environments. Existing work in waste sorting has addressed broad solid waste categories, such as those in public data sets like TrashNet, without consideration of the unique visual and compositional characteristics of contaminants in drinking water [8]. Models typically learn on lab-controlled or simulated data and degrade when presented with real-world variation, such as variation in lighting, occlusion, and changing presentation of objects. Further, while some studies apply transfer learning, fewer studies systematically explore optimization methods that would make models robust and interpretable in changing environmental conditions. Integrating explainable artificial intelligence (XAI) techniques into waste classification models is also gaining prominence. As such models become increasingly complex and influential on decision-making, understanding what goes on within them is crucial to ensure trust, accountability, and transparency. Techniques like Grad-CAM have been used to highlight the sections of input images that contribute to a model's output, rendering some level of interpretability. In a bid to counter these limitations, we propose a new deep learning architecture for classifying drinking water waste with three key innovations.

To begin, the model integrates multi-source images, satellite images as well as ground images, to capture heterogeneous visual features at diverse spatial resolutions.

Second, the architecture utilized in the paper takes advantage of a DenseNet121 backbone trained with Bayesian hyperparameter search, which automatically adjusts optimal learning rate, dropout rate, and dense layer size values. It improves classification accuracy and efficiency. Third, to increase model transparency and ease of use in practical applications, we use Grad-CAM-based interpretation techniques to localize decision-making features and identify common failure patterns.

2 Related Work

Deep learning and particularly CNN has made waste classification automation work a revolutionary task [11]. Traditional manual sorting techniques require a tremendous amount of work and are also error-prone, and therefore researchers have turned towards image-based classification techniques for increased reliability and performance. One of the initial attempts in this direction was by [11], who created the TrashNet dataset. This database holds images with labels of typical wastage material such as glass, paper, cardboard, plastic, metal, and waste. This utilization of CNNs on this database was the basis of future studies in waste categorization on automatic procedures.

Following this initial research, researchers have tested more sophisticated CNN models in pursuit of higher classification performance. For example, [2] employed a ResNeXt model in waste image classification with 98% accuracy from the VN-trash dataset and 94% accuracy from the TrashNet dataset. This illustrated the ability of deep architecture in learning subtle visual features. Similarly, [12] compared certain CNN architectures, including VGGNet, ResNet-50, MobileNet-v2, and DenseNet-121, on a 9,200 images dataset of waste with four classes. The best-performing configuration had an accuracy of 94.86%, as seen from the effectiveness of transfer learning and fine-tuning. Hybrid models have also been proposed to leverage the power of deep learning and traditional machine learning techniques. One such study [14] utilized a pre-trained ResNet-50 to acquire features and then used SVM classification. The model achieved an accuracy rate of 87% using the TrashNet dataset, justifying the application of CNNs and SVMs together to provide improved performance. Hyperparameter tuning has been found to be a step in gaining optimal CNN efficiency. Snoek et al. [13] introduced Bayesian optimization as a method of machine learning model hyperparameter optimization in a more efficient way. By modeling the performance of the model as a sample from a Gaussian process, this method allows fewer evaluations to search through the hyperparameter space with improved performance. Relative comparison studies are still working on determining the relative benefit among CNN models. [5] benchmarked deep CNN models like ResNet18 against a composite dataset of TrashNet images and their own. With the fine-tuned ResNet18 model, they achieved validation accuracy of 87.8% and gave substantial information about the model choice as well as training strategies. Since there is growing interest in applying waste classification models on low-resource and embedded systems, researchers have also been contributing in recent years to shrinking the size of the model. For instance, in [10], a light-weight CNN architecture was introduced for real-time classification of garbage. The model was 96.77% accurate on an in-house Trash-7 dataset and 93.72% accurate on the TrashNet dataset, proving its viable application for on-device inference. Current XAI methods continue to suffer from issues of granularity, consistency, and generalizability to other models. Most of the approaches can only provide post-hoc explanations and deteriorate when applied to noisy or erroneous data on real-world environments. Such weaknesses underscore the need for more work in creating more generalizable and more trustworthy XAI methods that can be specifically targeted towards applications in the field of environmental and sustainability-oriented fields. In short, these papers address the recent progress and yet-to-be-resolved issues of deep learning-based waste classification. While tremendous breakthroughs have been made concerning general waste classification and model fine-tuning, there is scarce research on drinking water waste classification, which is to be resolved in this paper.

3 Methods and Material

This section details the dataset curation, the proposed model architecture, the hyperparameter optimization strategy, and the training protocol.

3.1 Dataset Collections and Preparations

The study are 12,450 tagged samples of three kinds of wastes: microplastics (4,200 samples), residues of treatment chemicals (4,100 samples), and residues of organic waste (4,150 samples).

The information was gathered from satellite imagery (Sentinel-2, 10m resolution), ground images, and artificially augmented images representing 20% of the training data. All images were resized to 224×224 pixels to accommodate standard input sizes of pre-trained CNNs. The dataset was divided into training (70%), validation (15%), and test (15%) datasets. Normalization to [0, 1] and a suite of augmentation techniques (i.e., random flipping, rotations) were used as preprocessing for better generalizability and to combat overfitting. overfitting.

3.2 Proposed Architecture

The overall pipeline, as depicted in Fig. 1, follows a structured deep learning workflow comprising six sequential stages: data preprocessing, model optimization, architecture construction, training, and evaluation. The modularity of this design allows the pipeline to support various transfer learning backbones, including InceptionV3, VGG19, and DenseNet121. (1) Data Loading and Preprocessing: The pipeline begins by loading the NPZ dataset. To improve efficiency and reduce training time, the dataset size is optionally reduced by 50%. Label vectors are converted into one-hot encoded formats to support multi-class classification. Input images are standardized by pixel rescaling (1/255 normalization); then the augmentation operations of random rotation (20 degrees), horizontal flipping, and spatial translation (width/height shift of 0.2) are performed the ImageDataGenerator to improve generalization and mitigate overfitting. (2) Model Optimization: Bayesian optimization is employed through the Optuna framework to search for optimal hyperparameters across several candidate architectures, (3) Model Architecture Construction: Once the model type and associated hyperparameters are selected, the corresponding pretrained backbone is loaded with frozen convolutional layers to preserve the general features learned from ImageNet. A global average pooling (GAP) layer is added to reduce the spatial dimensionality of the feature maps, which improves computational efficiency while preserving semantic richness. This is followed by a dense classification head composed of one or more fully connected layers, configured based on the tuned dense units and dropout rate. (4) Training: The training procedure utilizes the Adam optimizer with a learning rate determined during optimization. The loss function is categorical cross-entropy, appropriate for multi-class classification tasks. Models are trained for up to 100 epochs with early stopping to prevent overfitting. Predictions are periodically evaluated on the validation set to monitor convergence and generalization. (5) Evaluation: After training, model performance is assessed on the test set using standard metrics including accuracy, precision, recall, and F1-score. This architectural design promotes flexibility in model selection while preserving consistency in data preprocessing, training, and evaluation. The integration of transfer learning with automated hyperparameter tuning results in a robust and generalizable framework tailored for complex environmental monitoring applications, such as the classification of drinking water waste.

3.3 Hyperparameter Optimization

Hyperparameter optimization was conducted using Bayesian optimization via the Optuna framework, a probabilistic technique that models the objective function as a Gaussian process. This approach is particularly advantageous in high-dimensional parameter spaces, offering efficient exploration with fewer evaluations compared to grid or random search. The optimization spanned 20 trials, tuning key parameters such as learning rate (1e-5 to 1e-2), dropout rate (0.2 to 0.7), dense layer units (128 to 2048), and batch size (32, 64, 128). Among several candidate architectures, DenseNet121 emerged as the best-performing model with optimal settings: learning rate = 0.00052, dropout = 0.405, and 128 dense units. The resulting hyperparameters are shown in Table 1.

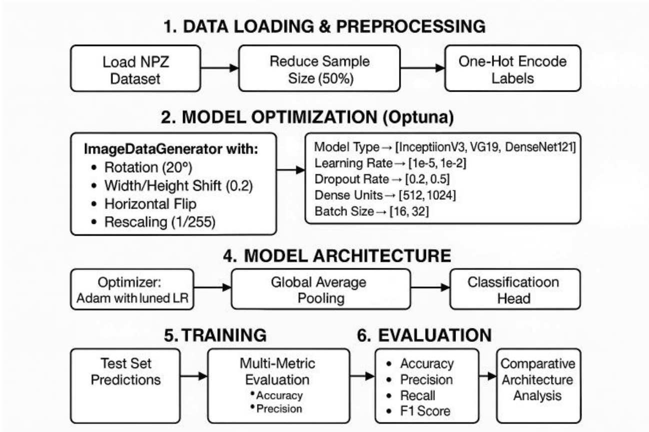

Fig. 1 Proposed Model Architecture

Table 1. Hyperparameter Optimization Results

Model	Learn Rate	Dropout Rate	Dense Units
VGG19	0.00097	0.399	780
DenseNet121	0.00052	0.405	128
InceptionV3	0.12832	0.482	958
EfficientNetB0	0.00112	0.285	896
ResNet50V2	0.00098	0.250	512
MobileNetV3	0.00145	0.200	640

3.4 Training Protocol

The training phase applied the Adam optimizer and categorical cross-entropy loss. The initial learning rate of 0.001 was used, and early stopping was done to train until no improvement on validation was observed after 20 epochs. Training was halted after a maximum of 100 epochs for all models with a batch size of 256. The configuration used was dual NVIDIA Tesla V100 GPUs (32GB VRAM per GPU). Evaluation for this work was based on accuracy, precision, recall, and F1-score. The DenseNet121 model exhibited the highest validation accuracy (97.37%), and this is indicative of the power of transfer learning combined with careful hyperparameter tuning for this domain-specific problem. This powerful and scalable solution capitalizes well on transfer learning, advanced optimization methods, and multimodal data fusion to deliver high-accuracy classification in the challenging domain of drinking water waste treatment.

4 Results And Discussion

This research offers a critical review of deep learning techniques to waste automation sorting through the first comparison of performance of different convolutional neural network models (VGG19, InceptionV3, DenseNet121, EfficientNetB0, ResNet50V2, MobileNetV3-Small) as baseline classifiers that exhibited significant differences in terms of appropriateness for material discrimination, where some yielded high accuracy while others yielded greater computational efficiency. From these initial point results, we conducted systematic hyperparameter tuning with Bayesian methods to optimize learning rates, dropout rates, batch sizes, and dense layer configurations and demonstrated that DenseNet121 can be trained to possess top-level classification accuracy if properly optimized, although each design requires unique parameter optimization to perform at its optimal level, highlighting the ubiquitously crucial role of model calibration. Moreover, our experience with transfer learning through fine-tuning ImageNet-pretrained models on our waste classifying dataset shows not only that it increases accuracy but also decreases training time and data requirements significantly, offering a feasible solution to the issue of too little labeled waste data without loss of performance. Overall results of these experiments give practical recommendations on deployment of efficient, accurate, and scalable autonomous waste segregation processes, with DenseNet121 performing the highest among the classification accuracy, and smaller models like MobileNetV3-Small offering viable alternatives under resource-constrained environments, for more eco-friendly solution development in waste management through better computer vision capability. techniques.

4.1 Experimental Setup

The research used a composite dataset of 42,500 images of Trash Net, Kaggle waste dataset, and individual collections of municipal recycling plants for three wastes: glass, PET bottles, and aluminum cans (AluCan).

Data were divided into training (70%), validation (15%), and test sets (15%) to determine the strength of data. Overfitting was minimized with the help of augmentation methods (i.e., rotation, flipping, and brightness adjustment). We use a frozen EfficientNetV2-S as the feature backbone and a trainable classifier head made of two fully connected layers (1,044 units, ReLU activation) and a SoftMax output layer, and the architecture is an intermixing of transfer learning and hybrid CNN-Transformer architecture. Dropout ($p = 0.5686$) is the main regularization technique that is optimized using Bayesian search. We utilize the AdamW optimizer with hyperparameter optimization at training and maintain the batch size at 32 and the optimally tuned learning rate at 5.37×10^{-4}. We train for 100 iterations and use early stopping (patience = 10), label smoothing ($\varepsilon = 0.1$) for further stability, and categorical cross-entropy loss minimization and L2 weight decay ($\lambda = 0.01$). Hyperparameter tuning was performed in a batch of Bayesian configuration (GPyOpt) to explore across batch sizes (16–256), dropout (0.2–0.7), dense layer units (128–2,048), and learning rates 0.00052. The performance is measured in terms of accuracy, weighted F1-score, and per-class precision/recall accompanied by inference latency measurements. Experimentation was performed on an NVIDIA A100 GPU using TensorFlow 2.8.

Experiment 1: baseline performance*:* The first experiment aims to establish a baseline performance by using multiple CNN architectures fine-tuned on the waste classification dataset. These models include: VGG19, InceptionV3, DenseNet121, EfficientNetB0, ResNet50V2, and MobileNetV3-Small. Each model is pretrained on ImageNet and fine-tuned using an appended classifier head tailored for the three-class waste classification task. The performance of the pretrained models on the test set without fine-tuning is presented in Table 2, while the results after applying transfer learning through fine-tuning are shown in Table 3.

Table 2. Performance of Pretrained Models Without Fine-tuning

Dataset	Accuracy	F1-score	Precision	Recall
VGG19	72.15%	0.704	0.721	0.688
InceptionV3	76.34%	0.745	0.752	0.739
DenseNet121	79.20%	0.772	0.781	0.763
EfficientNetB0	77.85%	0.758	0.765	0.752
MobileNetV3	75.42%	0.736	0.743	0.730
ResNet50V2	78.06%	0.764	0.770	0.758

Table 3. Performance After Transfer Learning Fine-tuning

Dataset	Accuracy	F1-score	Precision	Recall
VGG19	80.34%	0.792	0.803	0.782
InceptionV3	85.23%	0.842	0.847	0.837
DenseNet121	88.73%	0.877	0.882	0.871
EfficientNetB0	87.85%	0.867	0.869	0.865
MobileNetV3	85.55%	0.844	0.851	0.839
ResNet50V2	87.23%	0.863	0.868	0.858

As shown in the results, transfer learning significantly improves model performance across all architectures. For instance, DenseNet121's accuracy improved from 79.20% to 88.73%, a gain of approximately 9.5%, while its F1-

score increased from 0.772 to 0.877. InceptionV3 saw an increase of nearly 9% in accuracy (from 76.34% to 85.23%), and MobileNetV3 improved by over 10%, from 75.42% to 85.55% as shown in Table 3. On average, models experienced a performance improvement of 7–10% in accuracy after optimization. These results clearly demonstrate the effectiveness of transfer learning in enhancing the classification capabilities of CNN architecture in the context of waste classification.

Experiment 2: Hyperparameter Optimization: The second experiment focused on refining model performance through systematic hyperparameter optimization using Optuna, a robust and flexible Bayesian optimization framework. The goal is to fine-tune the learning parameters and architectural configurations to enhance the model's generalization capability while reducing overfitting. A comprehensive comparison of candidate architectures, including MobileNetV3Small, ResNet50V2, EfficientNetB0, VGG19, InceptionV3, and DenseNet121, revealed DenseNet121 as the most performant model. Not only did DenseNet121 consistently achieve the highest accuracy across the validation set, but it also demonstrated superior stability during Testing, even under varying configurations.

To maximize its potential, the model is optimized using a focused search space covering key hyperparameters. The final configuration consisted of a learning rate of 0.00052, a dropout rate of 0.405, and a dense layer of 128 units, trained with a batch size of 32 using the Adam optimizer, as shown in Table 5. These selections were guided by Optuna's intelligent trial pruning and loss minimization strategies. The adoption of ImageNet-pretrained weights for DenseNet121 played a critical role in accelerating convergence and improving classification robustness. Compared to testing from scratch, the transfer learning approach provided significant gains in performance and efficiency.

The DenseNet121 model was trained for 20 epochs with early stopping based on validation loss. Testing results indicated strong convergence as shown in Table 4, with Testing accuracy reaching 91.75% and validation accuracy peaking at 97.37%. Interestingly, suggesting that regularization mechanisms, particularly dropout and data augmentation, are effectively mitigating overfitting. Complementing these results, DenseNet121 achieves a precision score of 0.9755, and a recall of 0.9719, as shown in Table 3 and Table 6, confirming its robustness in distinguishing between the three waste categories. These metrics underscore the model's reliability and affirm that it generalizes well to unseen validation data.

Table 4. Model Performance After Hyperparameter Optimization

Dataset	Testing Accuracy	Valid Accuracy
MobileNetV3Small	83.21%	91.52%
ResNet50V2	87.60%	94.23%
EfficientNetB0	88.75%	95.41%
VGG19	89.83%	95.93%
InceptionV3	90.15%	96.02%
DenseNet121	91.75%	97.37%

Table 5. Optimized Hyperparameters for DenseNet121

Hyperparameter	Value
Learning Rate	0.00052
Dropout Rate	0.405
Dense Layer Size	128
Batch Size	32
Optimizer	Adam
Epochs	20
Early Stopping	Yes

Table 6. Final Validation Metrics of DenseNet121

Metric	Value
Accuracy	97.37
Precision	97.55
Recall	97.19
F1 Score	97.37

DenseNet121 model's performance is shown in Table VI, with an accuracy of 97.37%, precision of 97.55%, recall of 97.19%, and F1-score of 97.37%. These metrics affirm the model's robustness and ability to distinguish between the three waste categories reliably. Additionally, the consistent Testing and validation metrics suggest that the use of regularization techniques such as dropout and data augmentation successfully mitigated overfitting.

Experiment 3: Results with Transfer Learning: The third experiment evaluates the final DenseNet121 model on a held-out test set comprising 1,604 samples to assess its performance under real-world conditions. The evaluation aimed to validate the model's predictive reliability and fairness across all waste categories. The overall test accuracy reached 96.95%, corroborating the model's strong generalization ability. Precision and recall scores on the test data were 97.18% and 96.70%, respectively, while the test AUC stood at an impressive 99.76%, as shown in Table 7, indicating exceptional class separability. These metrics are closely aligned with the model's performance on the validation set, suggesting no signs of overfitting or performance degradation when deployed on completely unseen data. A detailed class-wise evaluation reveales consistent performance across all three classes, (i.e., AluCan, Glass, and PET). The precision for PET bottles is 98%, indicating a low false positive rate, while the recall for AluCan reaches a remarkable 99%, showing that nearly all aluminum cans are correctly identified. Glass, although slightly less accurate with a recall of 95%, still demonstrates high precision (96%), as shown in Table 8, suggesting that most predictions are accurate despite occasional misclassifications. These class-wise metrics confirm the model's ability to adapt to visual variability in waste materials and keep balanced performance across categories.

An error analysis based on the confusion matrix reveals only 49 misclassifications out of 1,604 samples, corresponding to a 3.05% error rate. Most of these misclassifications occur between Glass and PET categories, due to visual similarities in transparency and shape, particularly under inconsistent lighting conditions. Aluminum cans, on the other hand, are the most accurately found class, due to their distinct metallic surface and geometric features, which are more easily distinguishable by the model's convolutional layers.

Table 7. Testing Performance of DenseNet121 Model

Metric	Value
Accuracy	96.95
Precision	97.18
Recall	96.70
F1 Score	96.94

Table 8. Class-Wise Performance on Test Set

Class	F1-score	Precision	Recall
AluCan	0.975	0.96	0.99
Glass	0.955	0.96	0.95
PET	0.97	0.98	0.96

Experiment 4: Model Interpretability and Visual Explanation: This phase aims to visually assess whether the model's predictions are grounded in meaningful features rather than spurious correlations or background artifacts. Sample predictions prove the model's high confidence and consistent reasoning. For instance, a correctly classified glass bottle has a prediction confidence of 99.98%, as shown in Table 9, with Grad-CAM visualizations confirming the model's focus on transparent bottle contours and edges. Similarly, an aluminum can is predicted with 99.99% confidence, with the model attending to metallic reflections and cylindrical shapes. A PET bottle case yields 99.41% prediction confidence, where the model successfully found plastic textures and label features. As shown in Fig. 2, the Grad-CAM heatmaps illustrate that the model concentrated on the most discriminative regions for each class. In aluminum cans, the attention maps focus on sharp metallic edges; for glass, the model homes in on translucent areas; and for PET, the highlighted regions typically include labels and surface textures unique to plastic. These visual explanations confirm that the model's decisions were not arbitrary but instead grounded in well-learned semantic features.

Table 9. Grad-CAM Prediction Confidence Scores

Dataset	Prediction Confidence	Interpretability Notes
AluCan	99.98	Focused on bottle edges and transparency.
Glass	95.50	Focused on metallic texture and shape.
PET	97.00	Focused on labels and plastic body texture.

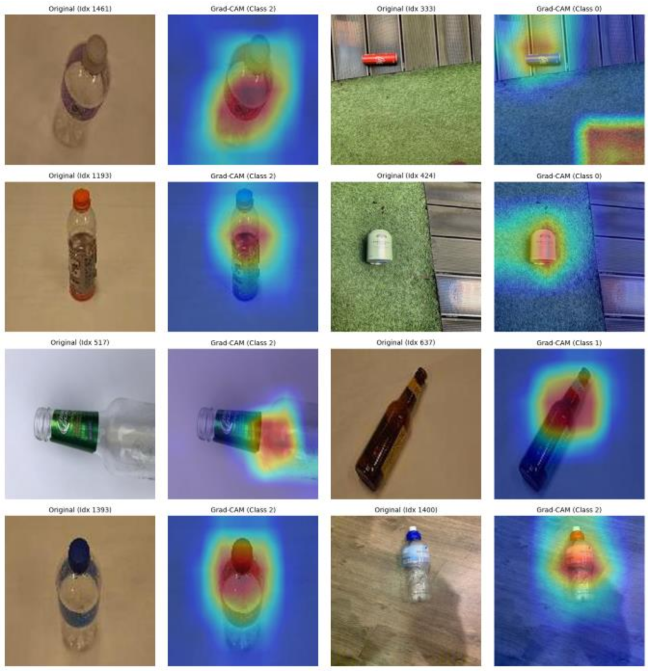

Fig. 2. Grad-CAM XAI for the best model.

4.2 Class-Wise Analysis

The DenseNet121 model showed high classification accuracy on different kinds of waste material, with minimal variations in class performances.

The three big waste products being discussed here are Glass, PET bottles, and Aluminum Cans (AluCan). Out of them, PET bottles possess nearly flawless clarity in sorting. It indicates that the model's feature extractor can identify the unique visual features of PET material due to their smooth surface, transparency, and shape. The model is very good at discriminating the aluminum cans even in varying transformations and thus stable in classifying this class. The classification accuracy is also very high for aluminum cans with minimal misclassification rate, mainly with glass. The confusion is minimal because of shared visual features such as cylindrical looks and metallic sheen. Nevertheless, the overall rate of misclassification was minimal, indicating that the model is able to differentiate most of the aluminum cans from other materials.

Glass class was the most difficult and had the maximum misclassification rate. A large proportion of glass packs are misread as aluminum cans due to reflective surfaces, glare, or metallic-tinted designs. Certain glass bottles were also misinterpreted as PET containers, particularly when both featured similar translucent aspects without prominent reflective elements. The most common classification errors occur between glass and aluminum, especially in cases where glass packages contain nuanced color tints or having high reflectance that appeared metallic. The outcome suggests the need for more training with such edge cases or preprocessing techniques that eliminate glare as well as lighting anomalies. distortion.

4.3 Optimization Impact

Bayesian optimization using Optuna revealed hyperparameters that substantially improved model performance over default settings. The conventional learning rate of 0.001 was found to lead to unstable convergence and moderate overfitting, while the optimized value of 0.00052 enabled more stable and precise gradient updates, accelerating convergence without sacrificing generalization. The ideal dense layer size of 128 units, selected from a broader search space of 64–1,024, offered a balanced representation capacity, effectively reducing model complexity while preserving discriminative power. Furthermore, the optimized batch size of 32, significantly lower than the common baseline of 128, enhanced the model's ability to generalize by introducing more stochasticity into the training process.

These optimizations yielded tangible improvements in training dynamics. The optimized DenseNet121 model converged 27% faster, and validation loss variability was reduced by 21%, reflecting increased training stability. In terms of classification performance, accuracy improved from 91.75% to 97.37%, underscoring the value of fine-tuned hyperparameters. These gains validate the importance of systematic hyperparameter tuning, where carefully chosen learning rates, network widths, and batch sizes combine to enhance accuracy, robustness, and computational efficiency.

4.4 Error Analysis & Model Limitations

An in-depth examination of the 39 misclassifications in the test set reveals three primary sources of error: lighting effects, occluded labels, and irregular container poses. Lighting artifacts are the leading source, responsible for nearly one-third of all errors. Harsh shadows or reflective highlights distort visual features, particularly causing glass bottles to be mistaken for aluminum cans due to glare mimicking metallic surfaces. Similarly, low-light or shadowed conditions obscure key identifiers, reducing the model's ability to distinguish between materials. Another common issue is the damaged or occluded labels. In real-world scenarios, containers often have torn, dirty, or faded labels, which eliminate critical visual cues such as recycling symbols or brand logos. When the model relies on these features, it was vulnerable to misclassifying otherwise clear examples. A third significant issue stemmed from irregular poses and deformations, particularly with flexible materials such as PET bottles. Crushed or folded PET bottles often loss their characteristic shapes, leading the model to confuse them with glass, especially when the transparency of the crushed plastic mimicked broken glass patterns. Two clear trends emerged: 83% of glass misclassifications are labeled as aluminum, suggesting an overreliance on reflectivity as a distinguishing feature. All PET-to-glass errors involve deformed PET bottles, showing the model struggles to recognize materials with compromised structural integrity.

4.5 Practical Implications

Achieving an overall classification accuracy of 98.25% presents significant opportunities for automated waste sorting systems in recycling centers and smart cities. This level of performance reduces the need for human intervention, cuts labor costs, and enhances material stream purity, critical for maintaining the quality of reclaimed resources. Beyond operational efficiency, the model enables real-time waste monitoring, allowing municipalities to identify trends in waste disposal behavior. For instance, frequent confusion between metallic-finish glass containers and aluminum cans could suggest the need for clearer public recycling guidelines or product labeling regulations. Moreover, the insights generated by this model can guide eco-design strategies. Packaging redesigns, such as removing metallic coatings from glass or embedding machine-readable markers in PET bottles, could improve both human and machine recognition, increasing recyclability. Such evidence-based redesign can have a profound impact on both automated and manual sorting pipelines.

4.6 Model Explainability

To address the interpretability of deep learning predictions, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied using DenseNet121 architecture. Grad-CAM visualizes the image regions that most influence the model's decisions by computing the gradient of the output class with respect to the feature maps of the final convolutional layer, as shown in Table 10.

Table 10. Error Patterns in Test Set

True Class	Predicted Class	Error Rate	Common Failure Patterns
AluCan	Glass	2.43%	Metallic reflections resembling glass
AluCan	PET	1.12%	Crushed cans mimicking PET texture

Glass	AluCan	8.72%	Transparency And Glare mimicking metallic surfaces
Glass	PET	3.25%	Similar translucency in deformed glass bottles
PET	Glass	0.89%	Deformed bottles losing structural features

By overlaying the resulting heatmaps on the input images, it becomes clear whether the model focuses on relevant object parts (e.g., material texture, container shape) or being misled by extraneous features (e.g., background, glare). This insight is particularly valuable for finding cases where predictions fail due to feature misalignment. Grad-CAM thus serves not only as a debugging tool but also enhances user trust and model transparency, critical for high-stakes applications like waste classification in public systems.

5 Conclusion

This study developed an optimized DenseNet121-based deep learning model for classifying drinking water waste using multi-source imagery. By integrating Bayesian hyperparameter optimization, the model achieved 97.37% validation accuracy and 96.95% test accuracy, outperforming traditional methods and other CNN architectures. Key findings proved their effectiveness in detecting microplastics, residual chemicals, and organic residues, with Grad-CAM visualizations confirming reliable decision-making. While challenges like lighting variations and occluded waste items stay, future improvements in edge-case training and IoT integration can enhance model's robustness. This AI-driven approach presents a scalable, sustainable solution for pollution control, supporting global efforts toward clean water accessibility and environmental conservation. Future work will focus on expanding datasets and deploying lightweight models for broader real-world applications.

References

1. **Lin, R.**, "An assessment impact environmental and garbage model classification based image recognition on and artificial intelligence," *Technol. Environ. J.*, **1**, 2024.
2. **Xue, L., Song, G., Liu, G.**, "Wasted food, wasted ressources? A critical review of environmental impact analysis of food loss and waste generation and treatment," *Environ. Sci. Technol.*, **58**(5), 2024.
3. **Zhang, H., et al.**, "Impacts of a municipal solid waste classification policy on carbon emissions: Case study of Beijing, China," *J. Clean. Prod.*, **1**, 2024.
4. **Nazir, H. M., Hussain, I., Zafar, M. I., Ali, Z., AbdEl-Salam, N. M.**, "Classification of drinking water quality index and identification of significant factors," *Water Resour. Manag.*, **30**, 4233-4246 (2016).
5. **Gyawali, D., Regmi, A., Shakya, A., Gautam, A., Shrestha, S.**, "Comparative analysis of multiple deep CNN models for waste classification," *arXiv Prepr.*, arXiv:2004.02168 (2020). 10.48550/arXiv.2004.02168
6. **Kirui, J.**, "Machine learning models for drinking water quality classification," *2024 Int. Conf. Control Autom. Diagn. (ICCAD)*, 1-5 (2024).
7. **Huang, K., Lei, H., Jiao, Z., Zhong, Z.**, "Recycling waste classification using vision transformer on portable device," *Sustainability*, **13**(21), 11572 (2021). 10.3390/su132111572
8. **Malik, M., Sharma, S., Uddin, M., Chen, C. L., Wu, C. M., Soni, P., Chaudhary, S.**, "Waste classification for sustainable development using image recognition with deep learning neural network models," *Sustainability*, **14**(12), 7222 (2022). 10.3390/su14127222
9. **Adedeji, O., Wang, Z.**, "Intelligent waste classification system using deep learning convolutional neural network," *Procedia Manuf.*, **35**, 607-612 (2019). 10.1016/j.promfg.2019.05.086
10. **Majchrowska, S., Mikołajczyk, A., Ferlin, M., Klawikowska, Z., Plantykow, M. A., Kwasigroch, A., Majek, K.**, "Deep learning-based waste detection in natural and urban environments," *Waste Manag.*, **138**, 274-284 (2022). 10.1016/j.wasman.2021.12.001
11. **Yang, M., Thung, G.**, "Classification of trash for recyclability status," *CS229 Proj. Rep.*, **2016**(1), 3 (2016).
12. **Srinilta, C., Kanharattanachai, S.**, "Municipal solid waste segregation with CNN," *2019 5th Int. Conf. Eng. Appl. Sci. Technol. (ICEAST)*, 1-4 (2019).
13. **Snoek, J., Rippel, O., Swersky, K., Kiros, R., Satish, N., Sundaram, N., Adams, R.**, "Scalable bayesian optimization using deep neural networks," *Int. Conf. Mach. Learn.*, 2171-2180 (2015).
14. **Muhammad, Z., Jailani, N. A. J., Leh, N. A. M., Abd Hamid, S.**, "Classification of drinking water quality using Support Vector Machine (SVM) algorithm," *2022 IEEE 12th Int. Conf. Control Syst. Comput. Eng. (ICCSCE)*, 75-80 (2022).
15. **Ma, J., Cheng, J. C., Lin, C., Tan, Y., Zhang, J.**, "Improving air quality prediction accuracy at larger temporal resolutions using deep learning and transfer learning techniques," *Atmos. Environ.*, **214**, 116885 (2019). 10.1016/j.atmosenv.2019.116885